

ABHINAV BIRRU

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Education

Indian Institute of Technology, Guwahati <i>Bachelor of Technology in Electronics and Communication Engineering</i>	Nov 2020 – May 2024 <i>CGPA: 8.67</i>
Telangana State Board of Intermediate Education <i>Senior Secondary</i>	June 2018 – Mar 2020 <i>Percentage: 98.5%</i>
Jawahar Navodaya Vidyalaya, Warangal <i>Secondary</i>	Mar 2018 <i>Percentage: 98%</i>

Experience

AQR Capital India Services LLP <i>Software Developer</i>	July 2024 - Present <i>Bengaluru, Karnataka</i>
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Historical Factor Refresh & Signal Library Consolidation

- Consolidated the research and production factor trees into a single unified tree, then drove the annual historical refresh for the resulting 250K+ factor library - recomputing every leaf-level RAW and derived signal back to 1980s - coordinating 2 teammates alongside.
- Designed a research-to-production backfill and stitching methodology for datasets like Betas, Universes whose production history is shorter, identified analogous research datasets, validates statistical equivalence, and stitched histories without introducing look-ahead.
- Reconciled divergent factor-computation logic between research and production code bases into a single canonical implementation adopted by both environments, removing a long-standing source of signal drift and reproducibility bugs.
- Extended the production factor tree to cover research-only signals, coordinating with signal owners to migrate, validate, and sign-off each factor.
- Diagnosed ~ 40 distinct failure modes across 600+ RAW factors (data-gaps, cross-environment logic differences) and built an owner-routed triage workflow to resolve the failures.

Infrastructure Modernization & Production Engineering

- Contributed to migration of a legacy Python 2 Windows analytics stack to a Python 3 Linux runtime on AWS Batch; refactoring the monolith into modular cloud-native libraries.
- Built and maintained Docker containers for reproducible execution across dev, research, and production environments.
- Migrated batch file storage from network shared drive pickle files to Parquet on AWS with S3 versioning, achieving ~ 70 – 80% reduction in storage footprint and eliminating a class of bugs caused by pickle protocol incompatibility between Python 3.6 and 3.11 environments; Parquet’s columnar format, embedded schema, and language-agnostic readability also made the data usable beyond python.

Research-Production Environment Consolidation

- Owned end-to-end consolidation of research and production across 3 workflows - data loading, universe updates, and factor computation/return loading - eliminating redundant research pipelines and redirecting analysts onto stable, signed-off production data.
- Stitched full research history onto the corresponding production datasets, establishing production as the single source of truth.
- Cut monthly research factor computation down to once-a-year annual refresh by routing day-to-day needs through production outputs, removing a recurring compute and support burden.
- Extended production factor-return loading to cover non-leaf research factors; migrated official research factor returns from SQL Server to Postgres and generalized the prod return-fetch API with an EnvironmentMode parameter to support research calls.

AQR Capital India Services LLP <i>Software Engineering Intern</i>	May 2023 - July 2023 <i>Bengaluru, Karnataka</i>
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Analytics Monitoring Dashboard (Python)

- Developed Python-based analytics library and Flask API for metrics ingestion into Elasticsearch. The library is used to capture and store metrics of running processes in Elasticsearch.
- Automated data collection from SQL and StoneBranch using time triggers in StoneBranch.
- Built Kibana dashboards for real-time process monitoring and analytics. Also developed a sophisticated dashboard using AngularJS for intuitive data visualization.
- Technologies used: **Python, Elasticsearch, AngularJS**

Projects

IITG-Campus-OLX  <i>Group Project, Coding Club, IITG</i>	Aug.2023 - Nov 2023
E-Commerce Web App Development: Designed and developed a React JS-based web application for buying, selling, and renting products within the IITG community. Firebase Integration: Implemented Firebase for user authentication and utilized Firebase Storage to securely store and manage product images. Elasticsearch for Product Search: Integrated Elasticsearch to store product details, enabling efficient and fast product search functionality for users. User Profiles & Product Management: Created user profiles to facilitate quick product listing, easy editing, and seamless management of user-specific activities.	
Library Management System  <i>Personal Project</i>	Jun.2022 - July 2022
Flask Web Application: Designed and implemented a Flask and MySQL-based web application for efficient library book management, catering to both users and librarians. Role-Based Interfaces: Developed separate, user-friendly interfaces for librarians (for book inventory management) and users (for book transactions and personalization). User Authentication & Features: Implemented secure user authentication and functionalities, enabling users to add, issue, reserve books, maintain personal bookshelves, and view transaction histories.	

Technical Skills

Languages: C, C++, SQL, Python, Java, JavaScript*
Tools/Frameworks: AWS, JupyterHub, ReactJS*, NodeJS*, Jenkins, Docker, Flask, NumPy, Pandas
Databases: MSSQL, ElasticSearch, Firebase
Relevant Course Work: Data Structures & Algorithms, Object Oriented Programming, Database Management System.
* *Elementary proficiency*

Achievements

Flipkart Grid 5.0: Among the Top 0.5% teams shortlisted in the Online quiz out of 3,97,375 teams
JEE Advanced 2020 Secured an All India Rank 4656 among 1,15,000 candidates.
JEE Mains 2020 Secured an All India Rank 651 among 11,20,000 candidates.

Leadership / Extracurricular

Department Placement Coordinator	May 2023 – July 2024
<i>Centre For Career Development</i>	<i>IIT Guwahati</i>
Responsible for managing placement process for EEE department.	